

Campaign Level

Budget pacing determines spending pressure

Campaign state

- Remaining budget
- Current pacing reference
- Time and delivery status

Pacing reference

(e.g., traffic-aware, PID, dual)

a_{it}^{ref}

Opportunity Level

Autobidder evaluates the current impression

Impression features

- User and context
- Ad and creative features
- Expected value / conversion

Autobidder

(e.g., learned bidder)

a_{it}^{raw}

Execution boundary κ

$$a_{it}^{exec} = clip(a_{it}^{raw}, (1 - \kappa) a_{it}^{ref}, (1 + \kappa) a_{it}^{ref})$$

Auction Environment

Many advertisers compete for each impression

Advertiser 1

Advertiser 2

...

Advertiser N

Real-time auction

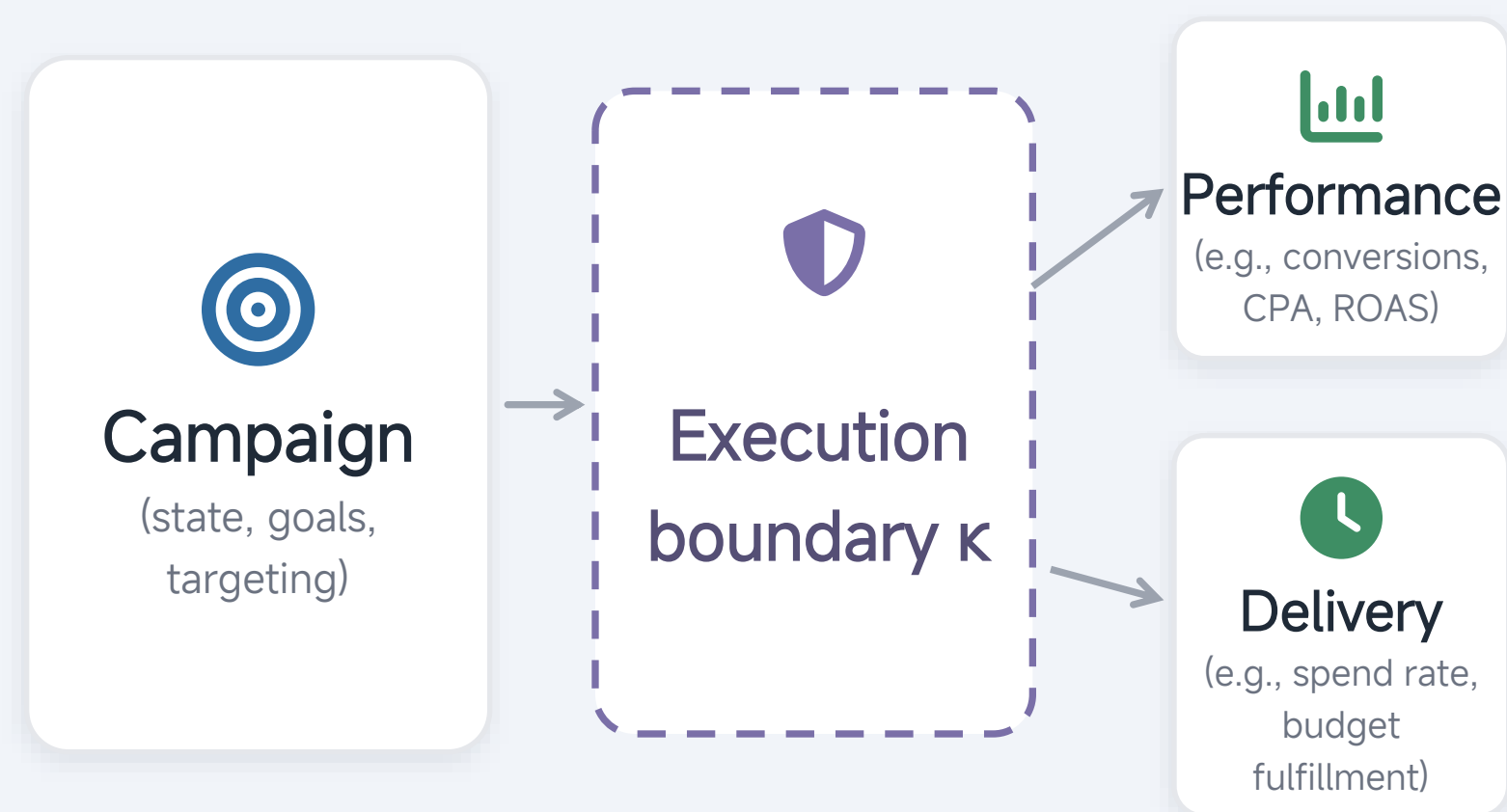
(e.g., second price)

a_{it}^{exec}

Winner

1. Campaign Performance-Delivery Frontier

How does execution discretion affect campaign outcomes?



2. Drivers of Discretion Value

When does wider discretion help?

Incremental selection information

Campaign selection room

Operating regime

3. Market Consequences

What happens when many advertisers adopt execution discretion?

Advertiser 1

Advertiser 2

...

Advertiser N

Shared marketplace

(real-time auction)

Ad allocation

Clearing prices

Platform revenue

Spillovers